Enrollment No.

# MID TERM EXAMINATION

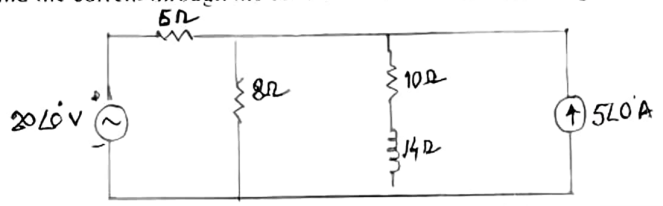
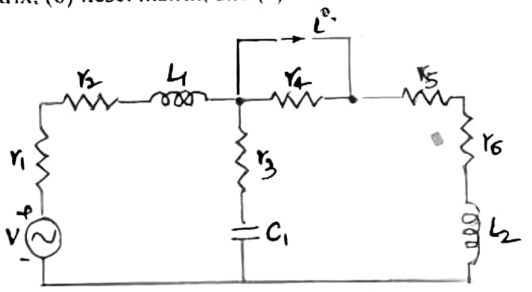
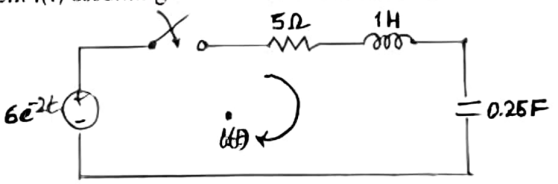
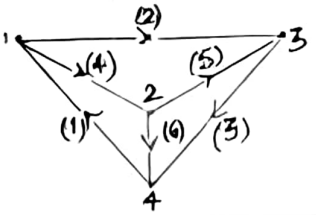
## B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Fourth Semester, March, 2025

Paper Code: EEC-206
Time: 1½ Hrs.

Subject: Network Analysis and Synthesis
Max. Marks: 30

**Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

| Q. No. | Questions | Max. Marks | CO(s) |
|---|---|---|---|
| 1 | (a) State Norton's Theorem | 3 | 1 |
| | (b) Define twigs and links of tree. | 2 | 1 |
| | (c) Find the Laplace transform of $e^{j\omega t}$, $t \geq a$ | 3 | 2 |
| | (d) Define transient and steady state Response. | 2 | 2 |
| 2 | (a) Find the current through the 8Ω resistor in the network of figure.   | 5 | 1 |
| | (b) For the circuit shown in figure, draw the oriented graph and write (a) incidence matrix, (b) tieset matrix, and (c) cutset matrix   | 5 | 1 |
| 3 | (a) Find the Laplace transform of $\int_0^t e^{-2t} t^3 \, dt$ | 5 | 2 |
| | (b) For the network shown in figure, the switch is closed at t = 0. Determine the current i(t) assuming zero initial conditions in the network elements.   | 5 | 2 |
| 4 | The graph of a network is shown in figure. Write the (a) incidence matrix, (b) tieset matrix, and (c) f-cutset matrix.   | 10 | 2 |